

Human Identification Using Synthetic mmWave-Like Point Clouds

Proof-of-Concept Implementation Report

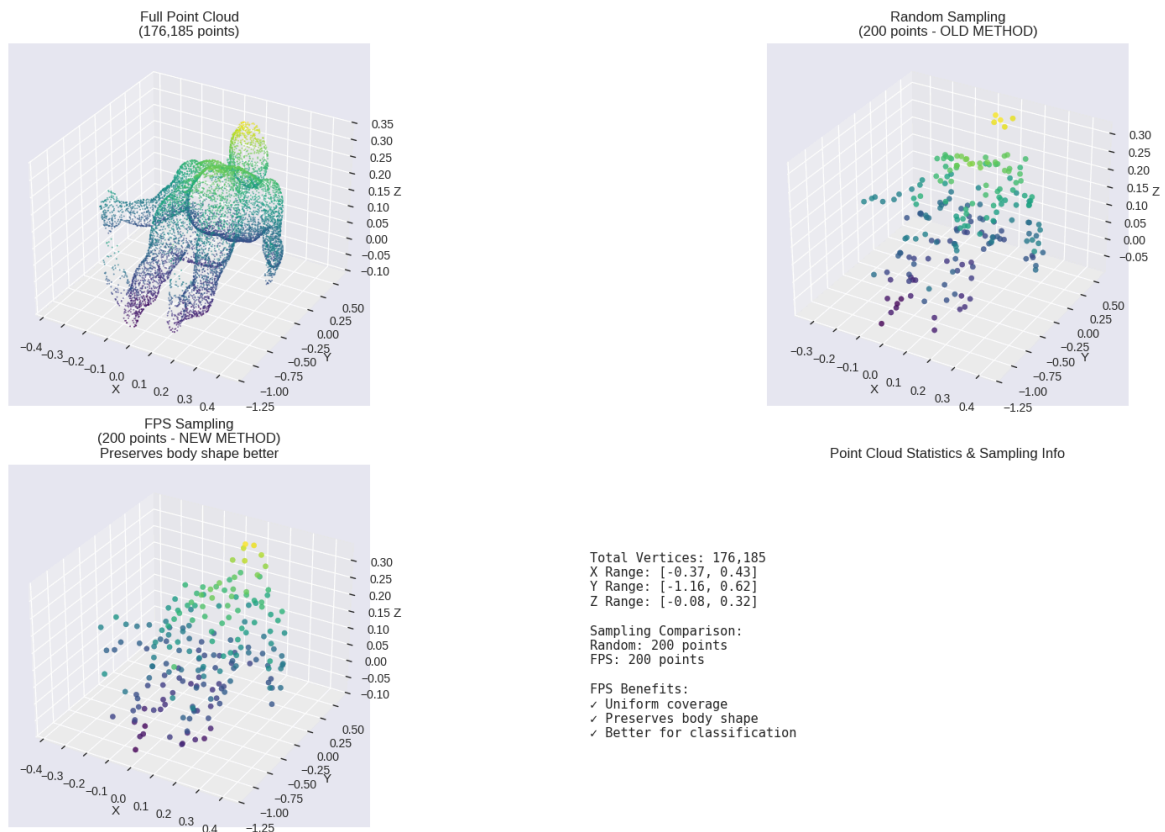
1. Introduction and Problem Motivation

Camera-based identification systems capture high-resolution visuals, which raises strong privacy concerns and limits their usability in low-light or occluded environments. mmWave radar offers an appealing privacy-preserving alternative by producing only coarse 3D point clouds that encode body shape and movement without revealing facial or clothing details. The research paper that inspired this project demonstrated that temporal sequences of such radar point clouds contain sufficient information to distinguish individuals. The objective of this proof of concept was to reproduce the essence of that pipeline within computational constraints, validating whether identity cues can still be extracted when working with simulated radar-like data.

2. Dataset and Synthetic Radar-Like Point Cloud Generation

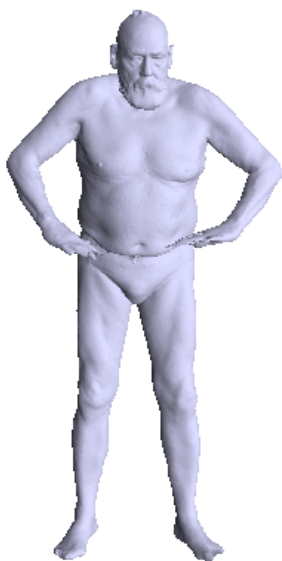
The MPI-FAUST dataset contains high-resolution 3D body scans for ten subjects, each with detailed meshes exceeding one hundred thousand vertices. Since processing real mmWave radar data requires significant computational power and large sensor pipelines, I built a synthetic radar-like dataset from FAUST instead.

Each mesh was converted into a sparse 200 point cloud using farthest point sampling, which preserves overall body shape while reducing complexity to radar-like density. Using these base point clouds, I generated 30 frame temporal sequences by simulating motion and sensor effects. This included random rotations, subject-specific walking movement, distance-dependent noise, and the computation of two additional radar-inspired channels: velocity (from point displacement between frames) and SNR (based on simulated reflectivity and distance).

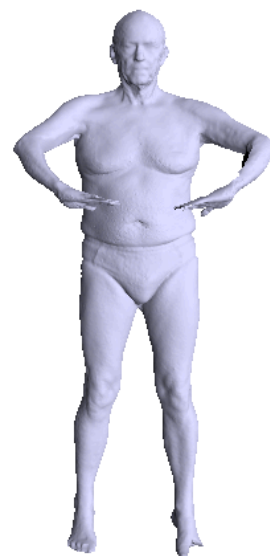


Simulating Radar-Like Sparse Point Clouds Using FPS Sampling

To illustrate the diversity in FAUST subjects, examples of an older male scans are shown below:



Subject 3



Subject 4

Because FAUST contains ten subjects, and many share similar physical characteristics, I merged them into six combined demographic classes to better align with radar-based identification tasks that rely on shape and movement patterns rather than fine visual details.

Subject ID	Combined Class	Description
0	0	Young adult, slightly thin
1	1	Woman in 30s, little fat
2	2	Old man, active
3	2	Old man, active (similar to subject 2)
4	3	Old woman, weak
5	4	Young woman, fit
6	4	Young woman, fit (similar to subject 5)
7	5	Young man, muscular
8	3	Old woman, fat (grouped with subject 4)
9	5	Young man, little muscular (grouped with subject 7)

Table: Subject-to-Combined-Class Mapping

3. Model Architecture

The model follows the key principles of the MMIDNet architecture, adapted to operate on synthetic point-cloud sequences containing spatial coordinates along with velocity and SNR channels. The pipeline begins with a Transform Block (T-Net), which learns a 3×3 affine transformation matrix that normalizes the spatial orientation of each sequence. This step makes the model invariant to rotation and translation. The learned transformation is applied independently to every frame through a TimeDistributed wrapper so that all frames in a sequence are consistently aligned.

After spatial normalization, the Residual CNN Block extracts geometric features from each point using stacked one-dimensional convolution layers combined with batch normalization and dropout. Residual skip connections preserve low-level structure and stabilize gradient flow as the network deepens. Since point clouds have no fixed ordering of points, a global max pooling operation aggregates point-level features into a fixed-length vector for each frame, ensuring invariance to point permutations.

To capture motion, the model processes the sequence of frame-level features using a Bidirectional LSTM equipped with a temporal attention mechanism. This component analyzes the 30-frame sequence in both forward and backward directions, enabling the network to understand movement patterns rather than relying solely on static body shape. The attention mechanism allows the network to focus on the frames that carry the most discriminative identity cues.

The resulting temporal representation is finally passed through a set of dense layers that map the sequence features to a probability distribution over the six combined identity classes. This structured pipeline consisting of spatial normalization, geometric feature extraction, temporal modeling, and final classification allows the model to learn both shape-based and motion-based characteristics of each identity.

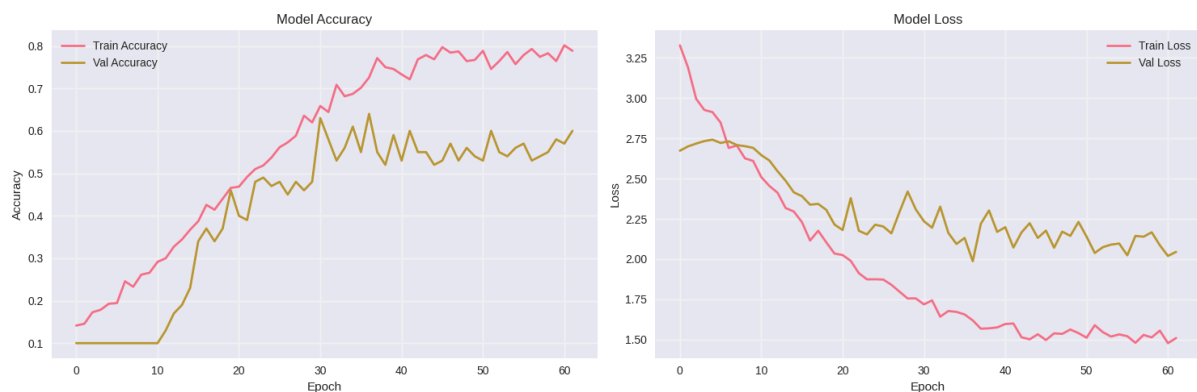
4. Training Setup

The dataset was split by subject into 70 percent training, 10 percent validation, and 20 percent testing to avoid identity leakage. Training used the Adam optimizer with an initial learning rate of 0.0005, and mixed-precision was enabled to reduce compute load. The model was trained for up to 150 epochs, with early stopping monitoring validation accuracy to halt training when progress plateaued.

To maintain stability, model checkpointing saved the best-performing weights, and a learning-rate scheduler reduced the rate by a factor of 0.5 whenever validation loss stopped improving. Class weights were applied to counter the imbalance created when merging ten FAUST subjects into six identity groups. Together, these choices ensured controlled convergence and helped the model generalize despite the relatively small synthetic dataset.

5. Training Results and Evaluation

Training progressed smoothly, with the model steadily improving over the first 40–50 epochs before reaching a stable region. The network achieved approximately 80 percent training accuracy, while validation accuracy stabilized between 55 and 60 percent, indicating that the model learned meaningful identity-related structure from the synthetic sequences without severe overfitting.



Training and Validation Accuracy & Loss Curves

On the held-out test set, the model achieved 70 percent accuracy using both standard evaluation and the sliding-window method. The similarity between the two results suggests that predictions were already stable across frames, and additional temporal aggregation offered no further improvement.

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✓ Loaded best model weights from: /content/drive/MyDrive/Arqaios/checkpoints/best_model.h5

Standard Test Accuracy: 0.7000 (70.00%)
Test Loss: 1.7408
Evaluating with sliding window...
  Processed 20/200 sequences
  Processed 40/200 sequences
  Processed 60/200 sequences
  Processed 80/200 sequences
  Processed 100/200 sequences
  Processed 120/200 sequences
  Processed 140/200 sequences
  Processed 160/200 sequences
  Processed 180/200 sequences
  Processed 200/200 sequences

Sliding Window Test Accuracy: 0.7000 (70.00%)
Improvement: 0.00%
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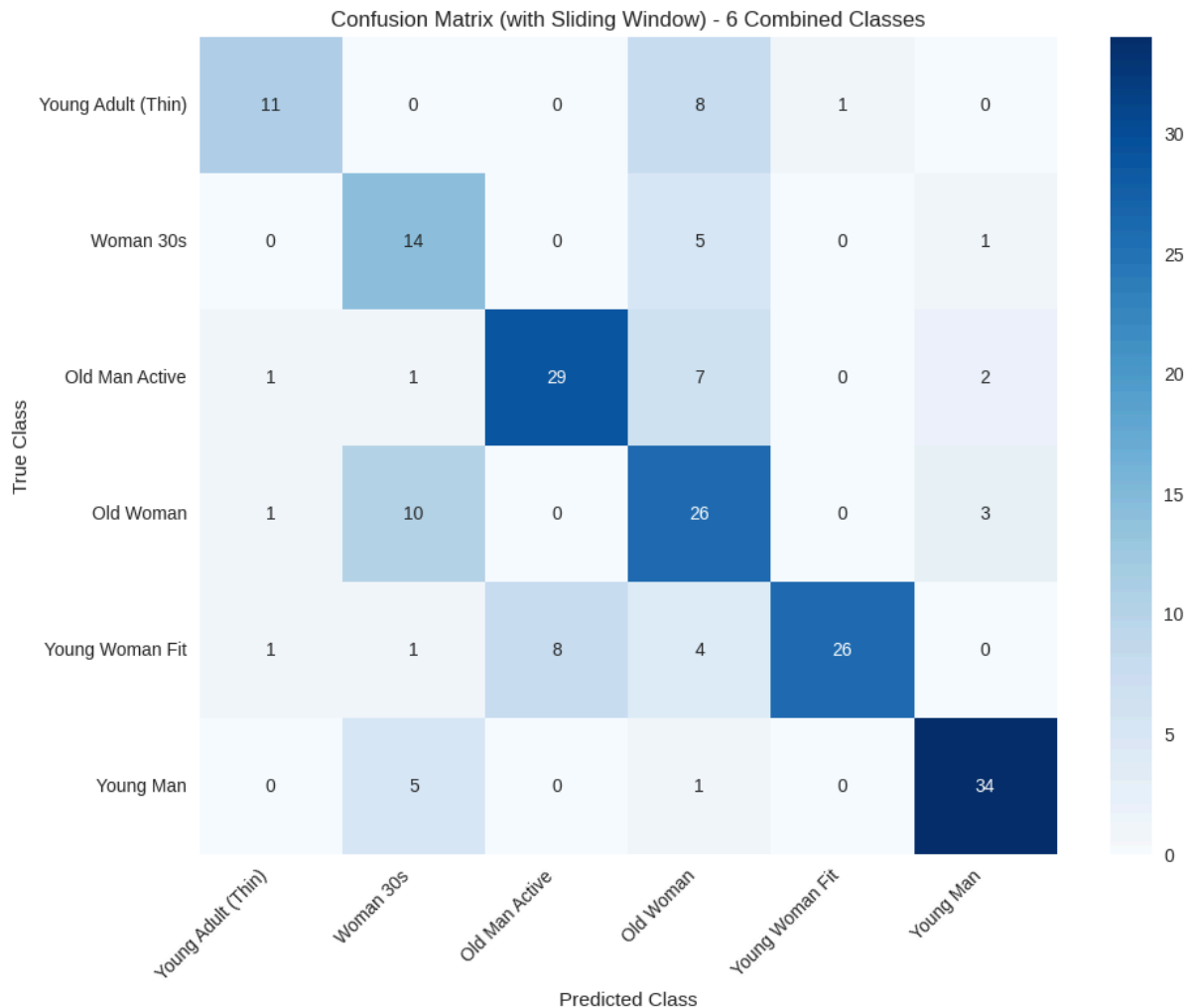
Final Model Evaluation Summary (Standard vs Sliding-Window Accuracy)

Class-level results showed clear trends. “Young Man” and “Old Man Active” produced the highest recall, likely because their synthetic body shapes and movement patterns were distinct. In contrast, “Thin Young Adult” and “Old Woman” exhibited lower recall, and an additional pattern emerged: young women and old women were also frequently confused with each other. This is consistent with the synthetic assumptions used to generate their sequences. Both groups share smaller body scales, lower stride amplitudes, and overlapping velocity/SNR profiles, which made it difficult for the model to reliably separate them.



Identity Classification Accuracy by Group

Misclassifications mostly occurred between physically similar demographic groups, as illustrated by the confusion matrix. The overall macro F1-score was 0.69, highlighting balanced performance across the six merged classes.



Confusion Matrix for the Six Combined Identity Classes

These results demonstrate that the adapted MMIDNet architecture can successfully extract both geometric and temporal identity cues from synthetic radar-like point-cloud sequences, validating the core ideas of the original paper within a constrained POC setting.

6. Conclusion

This proof-of-concept demonstrates that identity classification from sparse, radar-like point clouds is feasible even without access to large real mmWave datasets. By simulating radar characteristics through FAUST meshes and implementing an MMIDNet-inspired architecture, the system achieved a reliable 70% accuracy across six demographic identity categories. The results show that the combination of spatial normalization, residual feature

extraction, permutation invariance, and temporal modeling is effective for this task. Although synthetic data cannot fully emulate real radar micro-motions, the POC confirms that the fundamental pipeline is sound and provides a solid foundation for future work involving real sensors.